TED (10)- 5052 **Register No. ....**

Revision-2010 **Signature. ....**

MODEL QUESTION PAPER

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY- MARCH 2013

SUB: OPTOELECTRONIC DEVICES

(Common for ...)

[Maximun Marks : 100] **Time : 3 Hrs**

Part A

I. Answer all questions in one or two sentences. Each carries two marks.

1. List any two types of materials used in optical fibre construction.
2. Define the Aperture angle in light transmission.
3. Name any two applications of Laser.
4. Mention any two advantages of acousto optic modulators.
5. Write the bit rate of SONET STC-1/OC-1.

Part B

II. Answer any five questions. Each carries 6 marks.

a. Describe linear scattering losses in optic fibres.
b. Write note on photonic crystal fibre.
c. Describe how semiconductors are used for light detection.
d. Describe the population inversion mechanism in Laser.
e. Write brief note on optical circulators.
f. Describe the working of Beam splitters.
g. Write short note on fiber solitons.

Part C

Answer one full question from each unit. Each carries fifteen marks

Module- I

III. a). Explain about different materials used in optical fibre construction. and Draw the (10)
b). Mention the advantages of Multimode fibres. (5)

OR

IV. Explain the various losses occurs in Optic fibre communication. (15)

Module - II

V. a). Describe the different optical process in a semiconductor. (10)

b). Describe working principle of an LED. (5)

OR

VI. a) Discuss about absorption and radiation in semiconductors (8)

b) Show how an Avalanche photodiode helps in light detection. (7)

Module – III

VII. a) Describe the working of a wavelength division multiplexing couplers. (7)

b) Explain the working principle of Erbium-doped fiber amplifiers. (8)

OR

VIII. a) Explain the different types Splicing techniques. (8)

b) Describe the working of a Magneto optic modulator. (7)

Module IV

IX. a) Explain the functions of different layers in OSI model. (15)

OR

X. a) Explain the concept of Wavelength routing. (7)

b) Differentiate optical burst switching with packet switching. (8)

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